

NGC 6569 Spectroscopy and SED Analysis

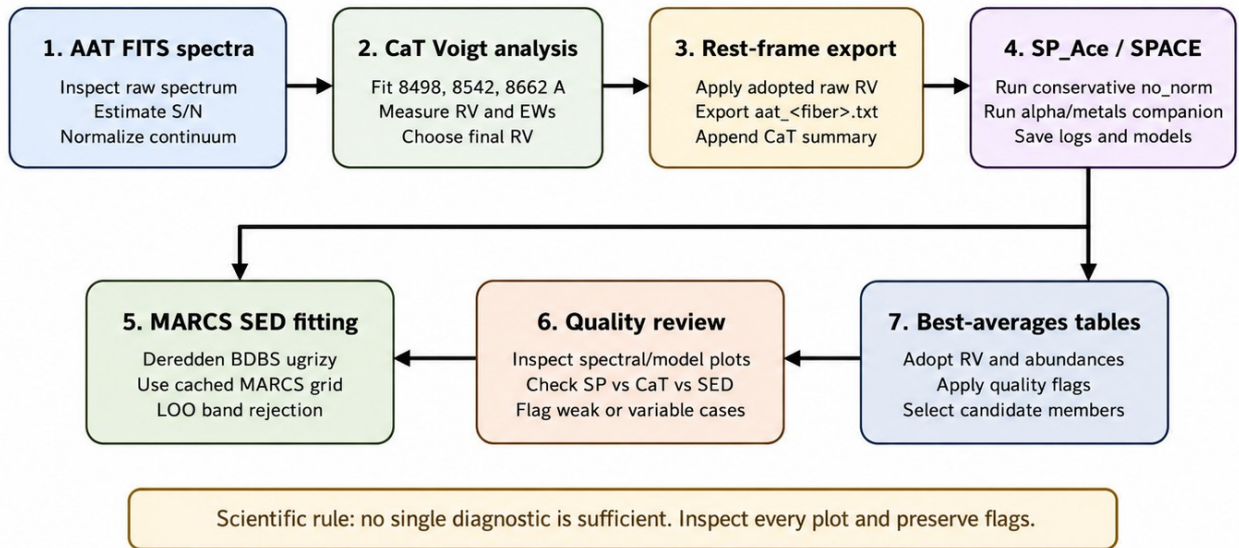
Master Student Help Manual

AAT/AAOmega Ca II Triplet Analysis, SP_Ace/SPACE, MARCS SED Fitting, and Construction of the Combined Best-Averages Tables

Prepared for students working on the MWBest NGC 6569 analysis

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NGC 6569 Analysis Workflow



Core scientific rule

Do not blindly trust an automated fit. Inspect every continuum, line profile, spectral model, and SED plot before accepting a result.

How to Use This Manual

This manual is the operational guide for reproducing the NGC 6569 results from the supplied Jupyter notebooks, shell scripts, descriptive PDFs, and final tables. It is written for students who may be new to the project but are expected to work carefully and preserve a reproducible audit trail.

The recommended approach is sequential: first reduce and measure each AAT spectrum, then run SP_Ace/SPACE, then perform MARCS SED checks, then review plots and construct or inspect the combined best-averages tables. Do not use the final tables as a substitute for understanding the measurements that produced them.

Important limitation of the uploaded package

The uploaded files document the workflow, but several operational dependencies are not included here: the raw n6569_2_*.fits spectra, the SPACE executable, the GCOGLibrarySPACE1 directory, the MARCS model grid, the Besançon comparison file, and several helper plotting scripts called by plotspectra_v1.ipynb. Students must obtain these from the project directory before running the complete pipeline.

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1. Scientific Purpose and Measurement Strategy

The Milky Way Bulge Extra-Tidal Star Survey (MWBest) investigates how inner-Galaxy globular clusters lose stars and contribute to the surrounding bulge field. For NGC 6569, wide-field BDBS photometry and Gaia astrometry were used to identify targets, and AAT/AAOmega spectroscopy in the Ca II triplet region supplied radial velocities, equivalent widths, and chemical information.

No single diagnostic is sufficient in the crowded and differentially reddened bulge. Proper motions alone overlap the bulge field; radial velocities admit contaminants; CaT metallicities require luminosity corrections; SP_Ace is limited by wavelength coverage and S/N; and SED fitting is sensitive to photometric quality and extinction. The pipeline therefore combines independent measurements and uses disagreements as quality information rather than hiding them.

1.1 The measurements used

Diagnostic	Primary role
BDBS ugrizy photometry	CMD position, dereddened colors, SED shape
Gaia astrometry	Proper motion, parallax, cross-identification
AAT CaT spectra	Radial velocity, S/N, CaT equivalent widths
CaT metallicity calibration	Independent [Fe/H] estimate from CaT2 and CaT3
SP_Ace/SPACE	Teff, log g, [M/H], [Fe/H], and where reliable [alpha/Fe]
MARCS SED fitting	Photometric consistency check and possible bad-band/variability flag
Combined tables	Adopted values, uncertainties, source flags, and candidate classifications

1.2 Why the red CaT spectra are used

The Ca II triplet lines at 8498.02, 8542.09, and 8662.14 Angstrom are strong in red giants and remain measurable in relatively faint stars. They are excellent for radial velocities and CaT-based metallicity calibration. The red spectral region is also less severely affected by interstellar extinction than the blue optical region.

The trade-off is that the CaT region contains relatively few independent alpha-element-sensitive lines. In this data set, [alpha/Fe] from the red spectra becomes substantially less reliable at modest S/N; approximately $S/N \geq 35$ is the practical threshold for trusting detailed alpha information without additional caution.

Why Blue Spectra Constrain $[\alpha/\text{Fe}]$ Better Than Red CaT Spectra

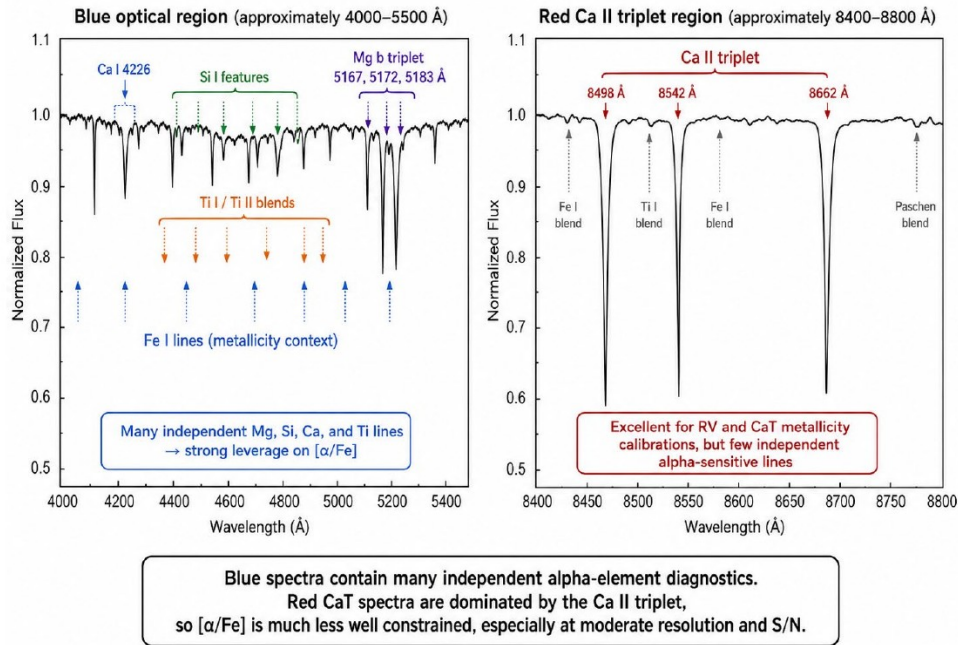


Figure 1. Blue spectra provide many independent alpha-element diagnostics, whereas the red CaT region is dominated by the three strong Ca II lines.

1.3 Current NGC 6569 selection context

- Adopted radial-velocity window: $-65 < \text{RV} < -25 \text{ km s}^{-1}$.
- Cluster-like metallicity window used in the current combined work: approximately $-1.1 < [\text{Fe}/\text{H}] < -0.5$ dex.
- The current strict adopted table contains 32 RV-plus-chemistry candidates.
- The broader paper-style Grade A-D reselection contains 37 candidates.
- These counts are version-specific to the June 2026 combined table and should be regenerated if any input measurements change.

Do not select members from one number alone

A plausible RV is not proof of membership. A plausible metallicity is not proof of membership. Candidate status is based on the combined kinematic, chemical, astrometric, spatial, and quality information.

2. Software, Directories, and File Organization

2.1 Recommended software

- Windows with Anaconda or Miniconda and Jupyter Notebook/Lab.
- Python packages: numpy, pandas, matplotlib, scipy, astropy, specutils, openpyxl, and pathlib (standard library).
- Windows Subsystem for Linux (WSL) or another Bash environment for the SPACE executable and .sh scripts.
- A PDF viewer and a spreadsheet program for reviewing diagnostic appendices and the final Excel workbook.

```
conda install numpy pandas matplotlib scipy astropy specutils openpyxl jupyter
```

2.2 Project directory layout

The notebooks and scripts assume the following directory structure. If your directory differs, edit the path variables at the top of each notebook or shell script before running it.

```

C:\Users\bella\Desktop\
|-- NGC6569_spectra\
|   |-- FITS_FILES\
|   |   |-- n6569_2_*.fits          raw AAT spectra
|   |   |-- aat_*.txt              normalized rest-frame spectra
|   |   |-- CaT_summary_table.txt  one-row-per-fiber CaT results
|   |   |-- CaT_summary_BDBS_XCSA0.csv merged CaT, BDBS, Gaia table
|   |   |-- plotting helper scripts and diagnostic outputs
|   |-- SPACE\
|   |   |-- SPACE                  executable
|   |   |-- GCOGLibrarySPACE1\    SP_Ace GCOG library
|   |   |-- run_*.sh              batch scripts
|   |   |-- space_model_*.dat      per-fiber fitted models
|   |   |-- SP_Ace summary tables
|   |   |-- MARCS\
|   |       |-- flx_wavelengths.vac[.gz]
|   |       |-- MARCS *.flx or *.flx.gz models
|   |       |-- MARCS_synthetic_ugrizy_cache.csv

```

2.3 Windows paths versus WSL paths

Python notebooks use Windows paths such as C:\Users\bella\Desktop\SPACE. Bash scripts running under WSL use the corresponding mounted path /mnt/c/Users/bella/Desktop/SPACE. Do not mix the two path styles within one environment.

Environment	Example
Windows/Python	C:\Users\bella\Desktop\SPACE
WSL/Bash	/mnt/c/Users/bella/Desktop/SPACE

2.4 Operational versus reference files

File	Use
AAT_SPEC_student_clean_voigt_v3.ipynb	Production one-star CaT reduction and SP_Ace export
run_best_conservative_no_norm_all_aat.sh	Primary SP_Ace/SPACE batch run without alpha mode
run_alpha_metals_best_conservative_no_norm_all_aat.sh	Companion alpha/metals run
SEDfit_v2.ipynb	MARCS setup, cache creation, single-star tests, and final batch SED run
plotspectra_v1.ipynb	Diagnostic plotting, appendix generation, TW plots, and sky/RV figures
Extra_Tidal_Stars_2025.pdf	Methods, current tables, diagnostics, and appendices
NewProject2026.pdf	Student-facing scientific overview and visual examples
Hughes_2026_AJ_171_137.pdf	Published NGC 6569 paper; Section 2.4.2 defines the Dias-Parisi CaT metallicity calibration.

Recommended file naming

Rename the uploaded shell scripts to remove the trailing “(1)” before use. This gives the operational names shown in the scripts and avoids awkward quoting at the command line.

3. Stage A: One-Star AAT CaT Analysis

Use `AAT_SPEC_student_clean_voigt_v3.ipynb` to process one FITS spectrum at a time. This notebook measures the CaT lines, supplies the adopted radial velocity, shifts the spectrum into the stellar rest frame, exports the `SP_Ace` input file, and appends the compact CaT results table.

3.1 Before starting a star

1. Run the notebook display/setup cell once after opening the notebook.
2. Run the “Clear old star variables” cell before every new star. This prevents measurements from the previous fiber from contaminating the next one.
3. Confirm that `spec_dir` points to the `FITS_FILES` directory.
4. Never alter or overwrite the raw FITS file.

3.2 Cell-by-cell workflow

Cell	Action	What to verify
0	Display and imports	Loads packages and plotting settings.
1	Clear variables	Deletes star-specific arrays, fits, EWs, RVs, and output variables.
2	Choose FITS file	Lists <code>n6569_2_*.fits</code> in numerical order and records the fiber ID.
3	Read FITS spectrum	Builds the linear wavelength solution from <code>CRVAL1</code> , <code>CDEL1/CD1_1</code> , and <code>CRPIX1</code> .
4	Inspect raw spectrum	Checks wavelength range, overall flux, CaT locations, defects, and missing data.
5	Estimate S/N	Uses robust scatter in pseudo-continuum windows near the CaT.
6	Full normalization	Fits a Chebyshev continuum in selected line-free windows.
7	Student quality decision	Requires a visual y/n decision on the normalization.
8	Restricted normalization	Optional CaT-only pseudo-continuum for cool stars or failed full-spectrum fits.
9	Initial line centers	Scans a common RV shift and finds local minima near the shifted triplet.
9b	Simultaneous Voigt RV	Fits all three lines with a shared RV over a broad search range.
10	Individual Voigt fits	Fits each line, integrates EWs, plots residuals, and combines line RVs robustly.

Cell	Action	What to verify
10b	RV comparison	Flags simultaneous-versus-individual disagreement larger than 10 km s^{-1} .
10c	Choose final RV	Student selects the physically credible Voigt or simultaneous solution.
11	Corrections and flags	Applies the observation-specific barycentric correction and uncertainty model.
12	Shift to rest frame	Uses the final adopted raw RV to move the normalized spectrum to lab wavelengths.
13	Export aat_<fiber>.txt	Writes 8450-8850 Angstrom rest-frame normalized flux for SP_Ace.
14	Append summary row	Appends RV, S/N, CaT EWs, and SumEW23 to CaT_summary_table.txt.

3.3 S/N estimate

The notebook estimates S/N using the median flux divided by a robust MAD-based scatter in selected pseudo-continuum windows. The value is a practical local S/N estimate for the CaT region, not a universal S/N for the entire spectrum.

Interpretation

Low S/N spectra may still yield useful RVs and CaT EWs. They are much less trustworthy for detailed SP_Ace alpha abundances.

3.4 Continuum normalization

The full-spectrum normalization fits a fourth-order Chebyshev polynomial to continuum windows that avoid the CaT cores and unstable red edge. The normalized continuum should remain close to 1.0 without a large slope, oscillation, or divergent red end.

- Good: continuum near 1.0, plausible broad line wings, no large-scale tilt, and no edge explosion.
- Bad: strongly sloped continuum, artificial bumps, polynomial divergence, or a cool-star pseudo-continuum that the polynomial cannot represent.
- When the full fit is bad, choose “n” and run the restricted CaT-only normalization. Do not force a visibly wrong full-spectrum fit.

Sky residuals are not removable stellar data

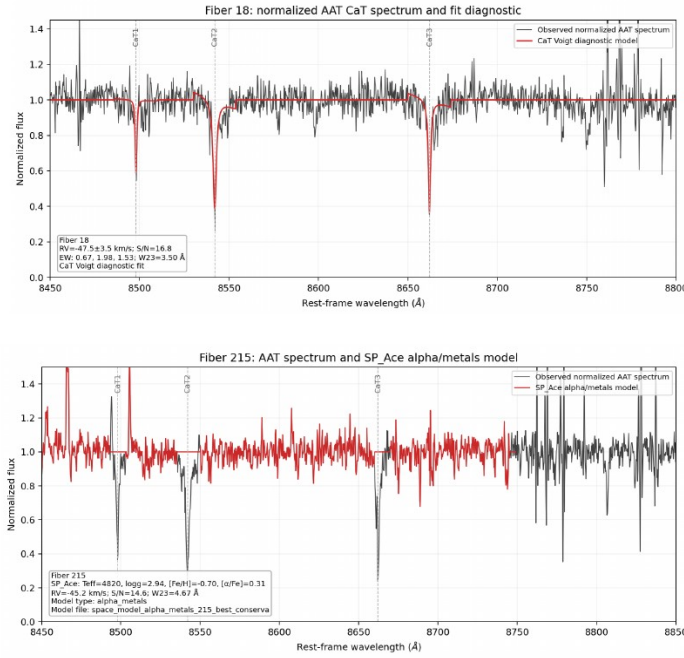
Residual night-sky emission, especially near the CaT region, can remain after reduction. Do not delete real pixels merely to make a fit look cleaner. Record the contamination and judge whether it compromises a line.

3.5 Voigt profiles, EWs, and RVs

The CaT lines have pressure-broadened wings and are represented more realistically by Voigt profiles than by simple Gaussians. The notebook fits a local absorption profile around each line and calculates the equivalent width by integrating the fitted absorption.

For non-relativistic stellar velocities, the line velocity is approximately $RV = c(\lambda_{\text{fit}} - \lambda_0)/\lambda_0$. The notebook compares the three line velocities and rejects inconsistent line

solutions when appropriate. All three CaT EWs are retained for the published Dias-Parisi calibration, while CaT2 + CaT3 is also stored as SumEW23 for two-line conversions and later robustness checks.



Normalized continuum, with Voigt fits to the CaT lines.

Normalized continuum, with SP_Ace fits to the regions outside the CaT lines, and avoiding residual sky emission lines.

Figure 2. Example normalized CaT Voigt fit and SP_Ace model comparison. CaT cores and sky-contaminated regions are not used in the conservative SP_Ace windows.

3.6 Dias-Parisi CaT metallicity [Fe/H]DP

The Dias-Parisi metallicity is calculated after the CaT measurement table has been merged with BDBS photometry, because the calibration requires both the CaT equivalent widths and the stellar apparent i-band luminosity relative to the horizontal branch. It provides an independent metallicity estimate and a fallback when SP_Ace is weak or unavailable; it is not a replacement for inspecting the spectrum.

Quantity	Value or source	Rule
EW8498, EW8542, EW8662	CaT1, CaT2, CaT3 from the Voigt fits	Use Angstrom units.
i	BDBS imag	Use the raw observed magnitude, not i0 = imag - Ai.
i_HB	16.59 mag for NGC 6569	This is the raw apparent BDBS i-band HB level.
beta_i	0.62	Published formal uncertainty 0.06; calibration scatter 0.17.

First calculate the three-line CaT sum and the luminosity-corrected reduced equivalent width:

```

SigmaEW_3L = EW8498 + EW8542 + EW8662
Wprime_i   = SigmaEW_3L + beta_i * (imag_raw - i_HB)

# NGC 6569 constants
beta_i = 0.62
i_HB   = 16.59

```


This is algebraically identical to the form used in the paper $\text{SigmaEW} = \text{Wprime}_i - \beta_i(i - i_{\text{HB}})$. A giant brighter than the HB has $i - i_{\text{HB}} < 0$, so the luminosity correction reduces the raw summed EW to Wprime_i .

Convert the reduced EW to the Dias-Parisi iron abundance with:

$$[\text{Fe}/\text{H}]_{\text{DP}} = -2.966 + 0.362 * \text{wprime}_i$$

Critical sign check

The constant is negative: use -2.966 in the calculation. A positive 2.966 would produce physically impossible, highly metal-rich values for these stars and indicates a transcription or typesetting error.

If only the two stronger red CaT lines are used, first convert their sum to the three-line-equivalent scale:

$$\begin{aligned} \text{SigmaEW}_{2\text{L}} &= \text{EW}_{8542} + \text{EW}_{8662} \\ \text{SigmaEW}_{3\text{L}} &= 1.26 + 1.00 * \text{SigmaEW}_{2\text{L}} \end{aligned}$$

The published coefficient uncertainties are 0.13 on the intercept and 0.03 on the two-line slope. The current combined tables also retain line-by-line W2 and W3 estimates, their average, their standard deviation, and a disagreement flag. Inspect a large W3-minus-W2 difference rather than accepting the average automatically.

```
def dias_parisi_feh(cat1, cat2, cat3, imag_raw,
                    i_hb=16.59, beta_i=0.62):
    sum_ew_3l = cat1 + cat2 + cat3
    wprime_i = sum_ew_3l + beta_i * (imag_raw - i_hb)
    feh_dp = -2.966 + 0.362 * wprime_i
    return wprime_i, feh_dp
```

Do not mix photometric systems

Use raw imag with raw $i_{\text{HB}} = 16.59$ mag. Do not subtract A_i from the star while leaving i_{HB} uncorrected. A consistently dereddened star and HB would give a very similar answer in this field, but mixing raw and dereddened quantities introduces a systematic error.

- The calibration is intended for luminous RGB stars and assumes the star is approximately at the distance of NGC 6569.
- For possible foreground/background stars, HB stars, RR Lyrae stars, very hot stars, or unusual AGB objects, treat $[\text{Fe}/\text{H}]_{\text{DP}}$ as uncertain and document the reason.
- Compare $[\text{Fe}/\text{H}]_{\text{DP}}$ with $[\text{Fe}/\text{H}]_{\text{SP}}$ when both are available. Agreement within about 0.30 dex is acceptable for an individual star in this data set; a larger difference requires a quality flag and inspection of the EWs, luminosity assumption, and SP_Ace fit.
- Never use a plausible $[\text{Fe}/\text{H}]_{\text{DP}}$ value to rescue an obviously bad CaT fit. The EW quality decision comes first.

3.7 Choosing the final RV

The notebook provides an individual-line Voigt solution and a simultaneous shared-RV solution. The student must inspect both. A mathematical convergence is not enough: the model may have locked onto a blend, residual sky line, or distorted profile.

- If both solutions agree and the line profiles are plausible, either should give a consistent result.
- If they differ by more than 10 km s^{-1} , the notebook sets a bad-RV-disagreement flag and the star requires manual review.
- Use the final selected raw/lab-frame RV for the rest-frame wavelength shift.
- Use the barycentric value for reporting and membership work.

3.8 Current RV corrections

For the current NGC 6569 observations, the notebook applies a barycentric correction of $-13.353688 \text{ km s}^{-1}$ and reports an empirical APOGEE comparison relation $RV_{\text{APOGEE}} = 0.999 RV_{\text{raw}} - 13.758 \text{ km s}^{-1}$. The adopted uncertainty combines the line-to-line scatter with a 3.204 km s^{-1} standards-fit scatter in quadrature.

Observation-specific value

Do not reuse the $-13.353688 \text{ km s}^{-1}$ barycentric correction for a different observing date, field, or target without recalculating it.

3.9 Required outputs for every star

- One exported rest-frame spectrum: `aat_<fiberid>.txt`.
- One appended row in `CaT_summary_table.txt`.
- Visual confirmation of the continuum and all three line fits.
- After the BDBS merge: raw imag, `Wprime_i`, `[Fe/H]DP`, and the CaT metallicity quality flag.
- A note for any sky contamination, blend, cool-star continuum, composite-looking spectrum, RV disagreement, or W2/W3 metallicity disagreement.
- No silent deletion of problematic stars: use flags and NaNs where appropriate.

4. Stage B: Batch SP_Ace/SPACE Fitting

SP_Ace/SPACE is run after the AAT notebook has produced normalized, rest-frame `aat_*.txt` spectra. The Bash scripts create a fresh `space.par` for every fiber, remove stale generic output files, run SPACE, save the exact per-fiber parameter file, preserve model and residual files, and append a master result table.

4.1 Prepare the WSL directory

```
cd /mnt/c/Users/bella/Desktop/SPACE
chmod +x SPACE
chmod +x run_best_conservative_no_norm_all_aat.sh
chmod +x run_alpha_metals_best_conservative_no_norm_all_aat.sh
```

The scripts expect the rest-frame spectra in `/mnt/c/Users/bella/Desktop/NGC6569_spectra/FITS_FILES` and match `aat_*.txt`.

4.2 Primary conservative run

```
bash run_best_conservative_no_norm_all_aat.sh
```

This is the primary atmospheric/metallicity run. Its adopted settings are:

- `fwhm` 0.8
- conservative wavelength windows: 8450-8493, 8503-8535, 8550-8660, and 8670-8745 Angstrom
- `no_norm` enabled because the spectra were already normalized in the AAT notebook
- `error_est` enabled
- `alpha` disabled
- `ABD_loop` enabled
- `Salaris_MH` enabled

The conservative windows avoid the CaT cores and the redder sky-heavy region while retaining nearby metal-line information. This setup was chosen because it gave the best agreement with the Voigt CaT and Dias-Parisi metallicity scale.

4.3 Alpha/metals companion run

```
bash run_alpha_metals_best_conservative_no_norm_all_aat.sh
```

This companion uses the same conservative wavelength windows but enables the alpha option. It produces alpha/metals model files and the master table `space_TGM_alpha_metals_best_conservative_no_norm_all_aat.txt`.

Alpha-abundance caution

The red CaT region contains limited independent alpha information. Treat $[\alpha/\text{Fe}]$ as reliable primarily for sufficiently high-S/N spectra (approximately $S/N \geq 35$), and always inspect the model plot.

4.4 What the scripts save

Output	Meaning
<code>space_<fiber>_best_conservative_no_norm.par</code>	Exact parameters used for the primary run
<code>space_TGM_ABD_<fiber>_best_conservative_no_norm.dat</code>	Per-fiber primary result
<code>space_model_<fiber>_best_conservative_no_norm.dat</code>	Per-fiber primary model spectrum
<code>space_residuals_<fiber>_best_conservative_no_norm.dat</code>	Per-fiber residuals
<code>space_<fiber>_alpha_metals_best_conservative_no_norm.par</code>	Exact alpha/metals parameters
<code>space_TGM_alpha_metals_<fiber>_best_conservative_no_norm.dat</code>	Per-fiber alpha/metals result
<code>space_model_alpha_metals_<fiber>_best_conservative_no_norm.dat</code>	Alpha/metals model used in plots
<code>run_*.log</code>	Complete console log and errors
<code>space_TGM_*_all_aat.txt</code>	Master tab-separated result table

4.5 How to monitor the run

```
tail -f run_best_conservative_no_norm_all_aat.log
# or
tail -f run_alpha_metals_best_conservative_no_norm_all_aat.log
```

The log should advance fiber by fiber. The script prints each input file, saves the per-fiber output, and reports warnings. The master output file should grow as rows are appended.

4.6 Failure handling

- If SPACE returns a non-zero status, the script records a warning in the log.
- If no usable result file is produced, the script writes a failed row with NaNs rather than silently dropping the fiber.
- The scripts delete old generic output files before each run so a stale result cannot be mistaken for the current star.
- The SP_Ace internal RV column is not the adopted science RV. The adopted RV comes from the CaT/XCSAO analysis and the barycentric value in the combined table.

5. Stage C: Spectral Plots and Publication Diagnostics

plotspectra_v1.ipynb is a launcher and plotting notebook. Several cells call external Python scripts, so first confirm that those scripts exist in FITS_FILES. Do not run every cell blindly; some cells are debugging or earlier alternatives.

5.1 Recommended production cells

Cell	Purpose
Cell 1	Run make_AAT_spectral_fit_pngs_and_appendix.py for CaT diagnostic PNGs/PDFs.
Cells 2-3	Run make_SP_Ace_bestfit_plots_and_appendix.py for standard model-fit appendices.
Cells 4-6	Debug a single alpha/metals model file and verify wavelength/flux columns.
Cell 7	Production alpha/metals plot run for all spectra; also makes PDF parts.
Cell 8	Build a 2x5-per-page appendix PDF and LaTeX include file from model PNGs.
Cell 9	Make one Tinsley-Wallerstein diagram for all or RV-selected fibers.
Cell 10	Make all-fiber and RV-selected TW diagrams using one shared RV color scale.
Cell 11	Make the sky-distribution plus RV-histogram figure.

5.2 Required visual checks

- Observed and model spectra use the same wavelength frame.
- The model columns were read correctly (normally wavelength column 0 and flux column 1).
- The CaT cores are excluded from conservative SP_Ace fitting rather than forcing the model through them.
- Residual sky emission is recognized and not interpreted as stellar abundance structure.
- The model continuum is sensibly aligned with the normalized observed spectrum.
- Tinsley-Wallerstein plots use the intended sample and preserve a shared color normalization when comparing panels.

5.3 Tinsley-Wallerstein diagram

The TW diagram plots $[\alpha/\text{Fe}]$ against $[\text{Fe}/\text{H}]$. The notebook can show all fibers or only the adopted RV-selected objects. The science points may be colored by barycentric RV or S/N and compared with the Besançon field model. An average uncertainty bar is placed in a clear corner of the plot.

Use the “best-rule” columns

For the current analysis, plots should use FeH_for_TW_rule and aFe_for_TW_rule (or the corresponding adopted columns in the final table), not an arbitrary single raw SP_Ace run.

6. Stage D: MARCS Photometric SED Fitting

SEDfit_v2.ipynb compares dereddened BDBS ugrizy photometry with MARCS model atmospheres. The SED is an independent consistency check on temperature, gravity, reddening behavior, and photometric quality. It is not a replacement for the adopted spectroscopic abundance scale.

6.1 Which SED notebook cells are current?

Cell	Use
Cell 0	Download/extract MARCS flux models. Run only when setting up the grid.
Cell 1	Repair or redownload flx_wavelengths.vac if an HTML page was saved by mistake.
Cells 2-3	Single-fiber nearest-MARCS demonstrations and diagnostic plotting.
Cell 4	Single-fiber restricted grid search using CMD plus SP_Ace priors.
Cell 5	Build MARCS_synthetic_ugrizy_cache.csv. Run once per MARCS grid.
Cells 6-8	Earlier batch and overlay versions retained for development/history.
Cell 9	Current production batch: restricted grid, leave-one-out rejection, and SP_Ace-nearest MARCS comparison.

Do not run all SED cells in order

The notebook contains developmental versions. For production, use the setup cells as needed, build the cache once, and run the final current batch cell.

6.2 MARCS wavelength file and the HTML error

A file beginning with <!DOCTYPE or <html is a web error page, not numerical wavelength data. Delete it and use the repair/download cell. The wavelength file is flx_wavelengths.vac or flx_wavelengths.vac.gz. The reader checks the gzip signature rather than trusting the filename extension.

6.3 Build the synthetic-photometry cache once

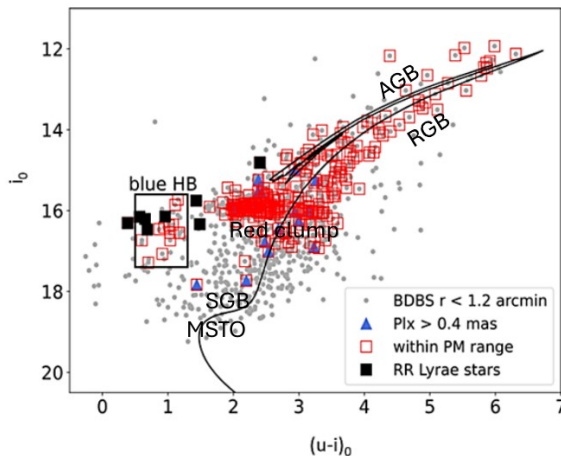
The cache-building cell reads every MARCS flux file, computes approximate top-hat synthetic nuFnu values in the BDBS-like ugrizy bands, and writes MARCS_synthetic_ugrizy_cache.csv. This is the slow step. Once the cache exists, each fiber can be fit in seconds because the code only fits scale and chi-square rather than rereading every compressed model.

6.4 Photometric preparation

1. Read BDBS magnitudes, magnitude errors, and extinction values A_u through A_y .
2. Deredden each band using $m_0 = m - A_{\lambda}$.
3. Convert AB magnitudes to flux density and then to νF_{ν} .
4. Add a 0.05-mag error floor in quadrature to the formal uncertainty.
5. Do not connect the broadband points with a line; they are independent filter measurements.

6.5 Restricted MARCS grid

The code first assigns a broad CMD evolutionary region from i_0 and $(u-i)_0$, then intersects that region with a local box around the SP_Ace Teff, $\log g$, $[\text{Fe}/\text{H}]$, and $[\alpha/\text{Fe}]$. Low-S/N spectra receive wider parameter boxes. Spherical models are preferred for lower-gravity stars and plane-parallel models for high-gravity stars when available.



The BDBS dereddened CMD for is shown for i vs. $(u-i)$, dereddened using the BDBS reddening maps (Simion et al. 2017). The small, gray, filled circles are the stars within 1.2' of the center of NGC 6569. The following objects are identified within 5 tidal radii of the center of the cluster: stars must have a proper motion within 1 mas/yr in R.A. and within 1.5 mas/yr in Dec of the main cluster proper motion: (~ 4.125 mas/yr, ~ 7.354 mas/yr). Additional constraints were applied: objects with BDBS u -magnitudes with uncertainties < 0.04 mag. were selected if they had detections in the BDBS g - and i -bands also. The red open squares were identified as targets for AAT/AAOmega spectroscopy. The blue filled triangles are flagged as likely foreground objects, and the black filled squares were identified as RR Lyrae stars. The isochrone shown is from the MIST models (Choi et al. 2016), scaled to be α -enhanced: $[\text{Fe}/\text{H}] = -0.75$ dex, $[\alpha/\text{Fe}] = +0.3$ dex, with an age of 13 Gyr.

Figure 3. Dereddened BDBS CMD and broad evolutionary regions used to guide the MARCS search. The red clump lies near the horizontal concentration around i_0 approximately 16 and $(u-i)_0$ approximately 2.5-3.

6.6 Scale and chi-square fit

For each cached MARCS model, only the multiplicative scale is fitted. With observed flux F_i , model band flux M_i , and uncertainty σ_i , the optimal scale is $s = \sum(M_i F_i / \sigma_i^2) / \sum(M_i^2 / \sigma_i^2)$. The model is ranked by chi-square = $\sum[(F_i - s M_i) / \sigma_i]^2$.

6.7 Leave-one-out photometry rejection

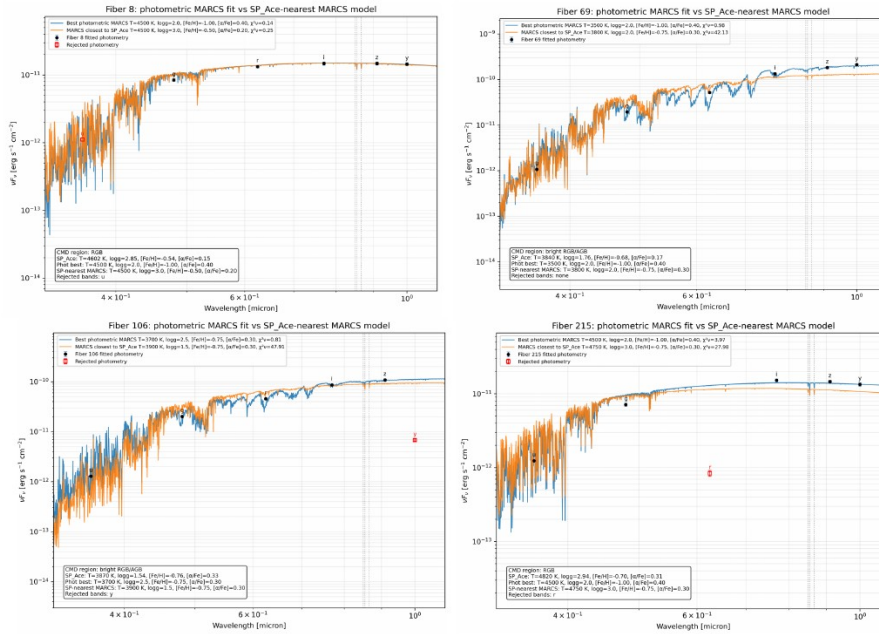
The current algorithm does not simply reject the largest residual from a compromised all-band fit. It removes each candidate band in turn, refits the remaining SED, and asks whether that omission produces a substantially better and physically smoother result.

- The omitted point must be at least 3 sigma from the leave-one-out refit.
- The reduced chi-square must improve by at least 1.0 or fall to 80 percent or less of the original value.
- Where two-sided neighboring bands exist, the point must create a local SED kink of at least 0.30 mag.
- At least four bands must remain.
- At most one band is automatically rejected in the current production setup.

Rejected points are plotted as open red squares. Accepted photometry is shown as filled black circles. The code also flags a possible variable star or broad photometric mismatch when multiple bands remain discrepant or the final reduced chi-square is large.

6.8 Interpreting the two MARCS curves

- Blue curve: the best photometric MARCS model within the restricted grid.
- Orange curve: the MARCS model closest to the SP_Ace atmospheric solution, independently scaled to the accepted photometry.
- Agreement supports the spectroscopic solution; disagreement prompts inspection but does not automatically replace SP_Ace.
- A single rejected band may reflect a bad measurement, non-simultaneous variability, blending, or residual extinction error.
- Multiple displaced bands can indicate variability or a broader model/reddening mismatch.



Sanity Check:

Photometry data fit to SED and spectral fits.

Figure 4. SED sanity checks compare the best photometric MARCS model with the MARCS model nearest to the SP_Ace solution. Red squares mark rejected bands.

6.9 Current SED outputs

```
C:\Users\bella\Desktop\SPACE\MARCS_SED_PLOTS_RESTRICTED_L00_REJECT_SPclosest_MARCS\
C:\Users\bella\Desktop\SPACE\MARCS_batch_restricted_fit_summary_L00_reject_SPclosest_MARCS.csv
```

The plots include the CMD region, SP_Ace parameters, best photometric model, nearest-SP_Ace MARCS model, reduced chi-square values, and rejected bands. The current display smoothing is 1800 logarithmic wavelength bins; this preserves visible spectral structure without plotting every native model sample.

7. Stage E: Combined Best-Averages Tables and Candidate Selection

The final products merge the CaT/BDBS/Gaia table, SP_Ace averages, CaT-DP metallicities, and MARCS/SED diagnostics using fiberid as the primary key. The adopted values preserve the strongest role of each method: CaT/XCSAO for RV, SP_Ace for atmospheric parameters where reliable, CaT-DP as an independent metallicity check/fallback, and SED as a consistency diagnostic.

7.1 Main delivered tables

File	Contents
NGC6569_full_machine_readable_best_averages.csv	335 rows, 86 columns; complete reproducibility table.
NGC6569_main_paper_best_averages.csv	214 rows; excludes table_use_flag = caution.
NGC6569_candidate_member_best_averages.csv	32 rows satisfying the current RV and chemistry conditions.
NGC6569_quality_summary_counts.csv	Counts by quality, source, and selection flag.
NGC6569_best_averages_tables.xlsx	Workbook containing README, main table, candidates, full table, quality summary, and column notes.
NGC6569_candidate_members_ranked_by_SNR.xlsx	37 paper-style candidates ranked by CaT S/N with model-file manifest.

7.2 Adopted-value logic

- RV_adopted: barycentric CaT/XCSAO velocity; this is the value used in the membership window.
- Teff_adopted and logg_adopted: currently from accepted SP_Ace averages.
- FeH_adopted: consensus/fallback logic using SP_Ace and CaT-DP; the FeH_source text explains the decision.
- aFe_adopted: currently SP_Ace where available; aFe_source records its provenance.
- SED_status, SED_ok, and SED_agreement: quality and consistency diagnostics, not quantities averaged into the adopted abundance.
- table_use_flag: primary, secondary, RV_CaT_only, or caution.

7.3 Current table selections

Selection	Rule	Current count
Full machine-readable	All uploaded spectra	335
Main paper	table_use_flag != caution	214
Strict candidate members	rv_member_window_NGC6569 = True AND chem_window_adopted = True	32
Strict candidates excluding caution	same, then exclude caution	26
Paper-style candidates	reapplied Grade A-D logic	37

7.4 Candidate-review principle

A table selection is a reproducible starting point, not a substitute for scientific review. Near-boundary cases, low-S/N stars, SED mismatches, and disagreements between SP_Ace and CaT-DP should remain visible through source and quality flags. The methods document identifies specific extended-RV, reinspection, watchlist, and rejected near-miss objects; students should not silently reclassify them without documenting the rationale.

8. Quality Assurance and Reproducibility Rules

8.1 Non-negotiable practices

1. Preserve raw FITS data unchanged.
2. Run the variable-clearing cell before every new spectrum.
3. Inspect every continuum and line fit; do not approve by convergence code alone.
4. Keep the exact per-fiber space.par files and run logs.
5. Use fiberid as the merge key and check for duplicates before merging.
6. Write failed measurements as NaN with a flag; do not drop them silently.
7. Keep adopted values separate from raw method-specific values.
8. When changing a threshold, rerun the full affected sample and record the new version/date.
9. Never hand-edit a final numerical result without recording the source and reason.

8.2 Minimum acceptance criteria by stage

Stage	Acceptance check
CaT normalization	Continuum near 1; no severe slope/divergence; CaT profiles physically plausible.
RV	At least two credible lines; no unexplained >10 km/s method disagreement.
Equivalent widths	Positive, plausible line areas; CaT2/CaT3 not dominated by defects.
SP_Ace	Model aligns with accepted windows; no stale file; chisq and parameters plausible.
Alpha abundance	High enough S/N or otherwise explicitly flagged as uncertain/lower-limit behavior.
SED	At least four accepted bands; rejected band documented; visual curve comparison performed.
Final table	Source flags populated; adopted values traceable; duplicate fiber IDs absent.

8.3 Suggested student processing log

Fiber	Normalization	RV choice	Sky/blend notes	SP quality	SED rejected band	Final flag
—	full / restricted	Voigt / simultaneous	_____	good / usable / weak	none / ____	_____
—	full / restricted	Voigt / simultaneous	_____	good / usable / weak	none / ____	_____
—	full / restricted	Voigt / simultaneous	_____	good / usable / weak	none / ____	_____

9. Troubleshooting

Problem	Action
No FITS files found	Check spec_dir, filename pattern n6569_2_*.fits, and Windows drive location.
Wrong wavelength scale	Inspect CRVAL1, CDELTA1/CD1_1, CRPIX1, and CUNIT1. Confirm CaT lines appear near 8498, 8542, and 8662 Å after the rest-frame shift.
Continuum fit is tilted or explodes	Use the restricted CaT-only pseudo-continuum; do not increase polynomial order blindly.
Automatic line center finds the wrong feature	Inspect the smoothed search plot, compare all three expected shifted positions, and use the simultaneous fit as a cross-check.
Voigt and simultaneous RVs disagree	Flag the star, inspect blends and sky residuals, and choose only a physically justified solution.
SPACE is not executable	Run chmod +x SPACE in WSL.
Shell script finds no aat_*.txt	Confirm Stage A exported the files into FITS_FILES and that the WSL path is correct.
SPACE produces no output	Read the log, inspect the per-fiber space.par, verify GCOGLibrarySPACE1, and check the input spectrum format.
A result looks copied from the previous star	The batch scripts remove generic outputs before each run; in the notebook, rerun the variable-clearing cell.
MARCS file gives <!DOCTYPE error	The file is HTML, not data. Delete it and run the repair/download cell.
gzip: not in gzip format	The file extension is misleading. The notebook checks the gzip signature and can read plain text; redownload if it is HTML or corrupted.
SED fitting is extremely slow	Build MARCS_synthetic_ugrizy_cache.csv once, then use the cached batch cell.
No models remain after restrictions	Check SP_Ace values, CMD region, MARCS grid coverage, and only then consider allow_fallback_wider_box = True.
Wrong photometric band rejected	Use the leave-one-out diagnostics. A single global-fit residual can be misleading because a bad band pulls the fit.
Several bands disagree	Do not reject multiple points automatically. Flag possible variability, blending, extinction mismatch, or model mismatch.
Legend or annotations cover data	Increase the log-y upper padding while keeping the legend inside the axes; do not move it over the title.

10. Student Completion Checklist

A student is finished with a fiber only when all applicable items below are complete.

☐ Raw spectrum inspected and fiber ID verified.

- ☐ S/N recorded.
- ☐ Continuum normalization accepted or restricted normalization used and documented.
- ☐ All three CaT fits inspected.
- ☐ Individual and simultaneous RV solutions compared.
- ☐ Final RV solution selected and barycentric RV recorded.
- ☐ CaT1, CaT2, CaT3, and SumEW23 recorded.
- ☐ Raw BDBS imag checked and Dias-Parisi [Fe/H]DP calculated with $i_{\text{HB}} = 16.59$ mag.
- ☐ W2/W3 metallicity agreement flag inspected.
- ☐ Rest-frame aat_<fiber>.txt exported.
- ☐ CaT_summary_table.txt row appended exactly once.
- ☐ Primary and alpha/metals SPACE runs completed or failure documented.
- ☐ SP_Ace model plot inspected.
- ☐ MARCS SED plot inspected; rejected band or mismatch documented.
- ☐ Final combined-table row checked against the source files.
- ☐ Any caution, variability, sky, blend, or model-disagreement flag entered in the processing log.

Definition of done

A numerical row without inspected plots and documented quality decisions is not a completed reduction.

Appendix A. Key Output Columns

Column	Meaning
fiberid	AAOmega/2dF fiber identifier; primary merge key.
RVraw	Raw/lab-frame CaT velocity before barycentric correction.
RV_adopted or RV	Adopted barycentric CaT/XCSAO velocity in km s^{-1} .
e_RV_adopted or e_RV	Adopted RV uncertainty in km s^{-1} .
SNR_CaT or S/N	Local CaT-region S/N estimate.
CaT1, CaT2, CaT3	Voigt-profile equivalent widths in Angstrom.
imag	Raw observed BDBS i magnitude used in the Dias-Parisi luminosity correction.
i0	Dereddened i magnitude used for CMD/SED work; do not substitute it for raw imag in the published DP calculation.
SumEW23	CaT2 + CaT3; retained for two-line conversion and robustness checks.
FeHDP_W2, FeHDP_W3	Line-specific CaT metallicity estimates used to diagnose inconsistent EWs.
FeHDP_avg_W2W3	Average CaT-derived metallicity from the W2 and W3 implementations when their agreement is acceptable.

Column	Meaning
FeHDP_flag	Quality flag for W2/W3 consistency and other DP-calibration problems.
Teff_avg_SP,logg_avg_SP	Averaged accepted SP_Ace atmospheric parameters.
FeH_avg_SP, aFe_avg_SP	Averaged accepted SP_Ace abundances.
SP_quality	good, usable, weak, or none.
SED_red_chi2	Reduced chi-square of the best restricted photometric model.
SED_auto_rejected_bands	Band removed by leave-one-out filtering, if any.
SED_agreement	consistent, check, or no_SED.
FeH_adopted	Final metallicity after consensus/fallback logic.
FeH_source	Text provenance of the adopted metallicity.
aFe_adopted	Adopted alpha abundance, currently from SP_Ace where available.
rv_member_window_NGC6569	True for $-65 < RV_{\text{adopted}} < -25 \text{ km s}^{-1}$.
chem_window_adopted	True for the adopted NGC 6569 metallicity window.
high_alpha_window	True for the adopted alpha-enhancement criterion.
table_use_flag	primary, secondary, RV_CaT_only, or caution.

Appendix B. Command Quick Reference

```
# Start Jupyter from Anaconda Prompt
jupyter notebook

# WSL: go to SPACE directory
cd /mnt/c/Users/bella/Desktop/SPACE

# Make executable
chmod +x SPACE run_best_conservative_no_norm_all_aat.sh \
  run_alpha_metals_best_conservative_no_norm_all_aat.sh

# Primary SPACE run
bash run_best_conservative_no_norm_all_aat.sh

# Alpha/metals SPACE run
bash run_alpha_metals_best_conservative_no_norm_all_aat.sh

# Monitor logs
tail -f run_best_conservative_no_norm_all_aat.log
tail -f run_alpha_metals_best_conservative_no_norm_all_aat.log

# Count exported spectra
ls /mnt/c/Users/bella/Desktop/NGC6569_spectra/FITS_FILES/aat_*.txt | wc -l

# Count saved alpha/metals models
ls space_model_alpha_metals_*_best_conservative_no_norm.dat | wc -l
```

Appendix C. Glossary and Essential Equations

Term	Definition
Stellar parallax	The apparent yearly displacement of a nearby star against distant background objects as Earth orbits the Sun. Distance in parsecs is $d = 1/p$ when p is in arcseconds.
Proper motion	The apparent angular motion across the sky, normally given as components in right ascension and declination in mas yr^{-1} .
Radial velocity	The line-of-sight component of motion, measured from the Doppler shift of spectral lines. Positive normally indicates recession and negative approach.
Equivalent width	The width of a rectangle at continuum level containing the same missing flux as an absorption line: $EW = \int (1 - F/F_c) d\lambda$.
Reduced CaT EW	For NGC 6569, $W_{\text{prime}_i} = \text{SigmaEW_3L} + 0.62(\text{imag} - 16.59)$, using the raw BDBS i magnitude.
[Fe/H]DP	Dias-Parisi CaT metallicity: $[\text{Fe}/\text{H}]_{\text{DP}} = -2.966 + 0.362 W_{\text{prime}_i}$. It assumes an RGB star approximately at the cluster distance.
Metal	In astronomy, any element heavier than helium.
Alpha elements	Elements mainly built by alpha-particle capture, including O, Ne, Mg, Si, S, Ar, Ca, and Ti.
[Fe/H]	$\log_{10}(N_{\text{Fe}}/N_{\text{H}})_{\text{star}} - \log_{10}(N_{\text{Fe}}/N_{\text{H}})_{\text{Sun}}$. A value of -1 means one-tenth solar iron abundance; -2 means one-hundredth.
[alpha/Fe]	The logarithmic alpha-element abundance relative to iron compared with the Sun.
CMD	Color-magnitude diagram; here commonly dereddened i_0 versus $(u-i)_0$.
RGB / AGB	Red-giant branch / asymptotic giant branch.
HB / red clump	Core-helium-burning phases. The red clump is the dense red horizontal-branch concentration.
MSTO / SGB	Main-sequence turnoff / subgiant branch.
Voigt profile	Convolution of Gaussian and Lorentzian broadening; useful for CaT lines with strong wings.
Barycentric correction	Correction from the observer's instantaneous motion to the Solar System barycenter.
Rest frame	Wavelength frame after removing the adopted stellar Doppler shift.
Reduced chi-square	Chi-square divided by the number of degrees of freedom; a diagnostic, not an automatic guarantee of correctness.

Appendix D. Source-File Inventory

File	Description	Role
AAT_SPEC_student_clean_voigt_v3.ipynb	One-star AAT CaT workflow and SP_Ace export.	Production
SEDfit_v2.ipynb	MARCS download/repair, cache, single-star tests, and final batch SED fit.	Production + history
plotspectra_v1.ipynb	Plotting launchers, appendices, TW diagrams, sky/RV figure.	Production + diagnostics
Extra_Tidal_Stars_2025.pdf	Detailed methods, reassessment, tables, and diagnostic appendices.	Reference
NewProject2026.pdf	Student overview, measurement definitions, figures, and scientific motivation.	Reference
Hughes_2026_AJ_171_137.pdf	Published NGC 6569 analysis; Section 2.4.2 gives the CaT/Dias-Parisi calibration and $i_{\text{HB}} = 16.59$ mag.	Reference
run_best_conservative_no_norm_all_aat.sh	Primary conservative SPACE batch script; rename without (1).	Production
run_alpha_metals_best_conservative_no_norm_all_aat.sh	Alpha/metals companion batch script; rename without (1).	Production
CaT_summary_table.txt	Compact 335-row CaT measurement table.	Intermediate
CaT_summary_BDBS_XCSAO.csv	335-row CaT plus BDBS/Gaia master input for SED/merge work.	Intermediate
NGC6569_full_machine_readable_best_averages.csv	Complete 335-row, 86-column combined output.	Final archive
NGC6569_main_paper_best_averages.csv	214-row manuscript-facing table.	Final paper
NGC6569_candidate_member_best_averages.csv	32 strict current candidates.	Final science subset
NGC6569_quality_summary_counts.csv	Quality and selection counts.	Final summary
NGC6569_best_averages_tables.xlsx	Multi-sheet combined workbook with README and column notes.	Final workbook
NGC6569_candidate_members_ranked_by_SNR.xlsx	37 paper-style candidates ranked by S/N plus model manifest.	Follow-up planning

This manual was assembled from the uploaded June 2026 project files. When the notebooks or selection logic are revised, update the manual version, output counts, and path examples together.